

ReDataX

M2: Cross-Market Multiscale Model

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1 Model role and testable hypothesis

M2 extends M1 with multiscale states from related markets. M1 describes one market in isolation and therefore cannot distinguish between a target-market imbalance that is reinforced by a neighboring market and one that is weakened by it. M2 sets out to describe several markets jointly and test whether this broader view improves predictive accuracy.

Hypothesis 1.1. After accounting for the multiscale state $X_A(t)$ of target market A , the synchronous states of the other markets $X_{S \setminus A}(t)$ contain additional information about aggressor-aligned markout. In other words,

$$\mathbb{P}(Y_{A,H}(t) = 1 \mid X_A(t), X_{S \setminus A}(t)) \neq \mathbb{P}(Y_{A,H}(t) = 1 \mid X_A(t)).$$

The M2-versus-M1 comparison follows the same controlled design as before: the information set $\mathcal{I}_t$ is the only intended difference. The target market, forecast horizon, admissible observations, classification algorithm, temporal splits, and metric calculations remain fixed. Any change in quality can therefore be attributed to the added cross-market information.

The implementation is named `rg_no_j`. The meaning of the `_no_j` suffix will become clear later in the document.

2 Why extend the model across markets?

BTCUSDT and ETHUSDT are not independent streams. Both markets use USDT as the quote currency, react to broad changes in liquidity, and may reflect the same impulse with different delays and amplitudes. This does not mean that one market determines the other. It does, however, leave room for a neighboring market to carry useful information after the target market's own multiscale state has already been accounted for.

The idea is motivated by market-microstructure research that links signed flow and order-flow imbalance to subsequent price changes [1, 3, 2].

3 Data source and common time grid

The source data are Binance Spot aggregate trades (`aggTrades`) for

$$\mathcal{S} = \{\text{BTCUSDT}, \text{ETHUSDT}, \text{ETHBTC}\}.$$

The ReDataX parser converts the `m` flag into the aggressor-side sign $\varepsilon_k \in \{-1, +1\}$, after which trades are aggregated into whole UTC seconds.

For every market $s \in \mathcal{S}$ and second t , the pipeline calculates aggressive buy volume $V_s^+(t)$, aggressive sell volume $V_s^-(t)$, and volume-weighted average price $P_s(t)$. If no trades occur during a second, both volumes are zero and the price remains undefined.

Definition 3.1. For market s during second t , the *flow field* is

$$\phi_s(t) = \begin{cases} \frac{V_s^+(t) - V_s^-(t)}{V_s^+(t) + V_s^-(t)}, & V_s^+(t) + V_s^-(t) > 0, \\ 0, & \text{when no trades occur.} \end{cases}$$

The field $\phi_s(t)$, whose range is $[-1, 1]$, describes the direction and imbalance of the trade flow.

Remark 3.1. ETHBTC is quoted in BTC, whereas BTCUSDT and ETHUSDT are quoted in USDT. This has no effect on the dimensionless ratio ϕ_s , but the current log-volume feature for ETHBTC starts in different units. Feature standardization is therefore essential.

4 Scale system and coarse-graining

ReDataX uses powers of two as its scale family:

$$\mathcal{B} = \{1, 2, 4, 8, 16, 32, 64\} \text{ seconds.}$$

For every market s and scale $B \in \mathcal{B}$, it computes the moving average over calendar seconds,

$$\bar{\phi}_s^{(B)}(t) = \frac{1}{B} \sum_{u=0}^{B-1} \phi_s(t-u),$$

including the current completed second. A full window is unavailable during the first $B-1$ seconds of the day, so the common sample begins no earlier than the 64th second. Seconds without trades remain in the denominator as zeros.

5 Target variable and observation mask

For target market $A \in \{\text{BTCUSDT}, \text{ETHUSDT}\}$, define the current orientation as $d_A(t) = \text{sign } \phi_A(t)$. The aggressor-aligned markout at horizon H , measured in basis points, is

$$m_{A,H}(t) = 10^4 d_A(t) \frac{P_A(t+H) - P_A(t)}{P_A(t)},$$

and the binary label equals one when this markout is positive:

$$Y_{A,H}(t) = \mathbf{1}\{m_{A,H}(t) > 0\}.$$

The final experiment uses $H = 5$ seconds.

The common M1/M2/M3 mask includes only times satisfying four conditions: at least 63 seconds have elapsed since the start of the day; the target market is active during second t and $d_A(t) \neq 0$; the target-market VWAP is finite and positive at both t and $t+H$; and $t+H$ remains within the same day. The related markets do not have to be active during the current second - zero flow is an admissible state for them.

6 M2 features and dimensionality

For every market $s \in \mathcal{S}$, scale $B \in \mathcal{B}$, and current target-market orientation $d_A(t)$, the model constructs four transformations of the field.

Definition 6.1. The *target-aligned flow* of market s with respect to market A is

$$h_{s,B}^{(A)}(t) = d_A(t) \bar{\phi}_s^{(B)}(t).$$

A positive value means that the flow of market s at scale B points in the current direction of target market A .

Definition 6.2. Imbalance amplitude and its even powers are defined as

$$r_{s,B}(t) = |\bar{\phi}_s^{(B)}(t)|, \quad a_{s,B}(t) = \left(\bar{\phi}_s^{(B)}(t)\right)^2, \quad b_{s,B}(t) = \left(\bar{\phi}_s^{(B)}(t)\right)^4.$$

These features allow a linear model to distinguish weak from strong imbalances.

The absolute size of the current second is recorded through the logarithm of total notional:

$$\ell_s(t) = \log(1 + V_s^+(t) + V_s^-(t)).$$

The complete M2 feature vector is

$$X_A^{\text{M2}}(t) = \left(\{\ell_s(t)\}_{s \in \mathcal{S}}, \{h_{s,B}^{(A)}, r_{s,B}, a_{s,B}, b_{s,B}\}_{s \in \mathcal{S}, B \in \mathcal{B}} \right),$$

with dimension

$$\dim X^{\text{M2}} = |\mathcal{S}| + 4|\mathcal{S}||\mathcal{B}| = 3 + 4 \cdot 3 \cdot 7 = 87.$$

7 Statistical model and training

Like the preceding models, M2 uses logistic regression. After fitting feature standardization on the training set, the probability of an adverse outcome is parameterized as

$$\hat{p}_A(t) = \sigma\left(\beta_0 + \sum_{s \in \mathcal{S}} \beta_s^\top z_s(t)\right), \quad \sigma(u) = \frac{1}{1 + e^{-u}}.$$

The parameters β are estimated by minimizing

$$\mathcal{L}(\beta) = - \sum_{i \in \text{train}} [Y_i \log p_i + (1 - Y_i) \log(1 - p_i)] + \lambda \|\beta\|_2^2.$$

The implementation uses `SGDClassifier` with logistic loss, L_2 regularization, a `StandardScaler`, averaged coefficients, and a fixed `random_state`.

Regularization matters for the same reason as in M1, and then some. Nested windows correlate features at neighboring scales, while BTCUSDT and ETHUSDT may react to common market factors, adding another source of collinearity.

The regularization parameter α is selected by mean daily AP on the development period from $\{10^{-5}, 10^{-4}, 10^{-3}\}$. M2, like M1, selects $\alpha = 10^{-3}$. Once the value is chosen, both the model and the scaler are refitted on the combined training and development samples.

8 Temporal validation and metrics

The main protocol does not shuffle individual observations across calendar periods. REAL-05 uses the following temporal split:

Training: 06.01.2025–19.01.2025,
 Development: 20.01.2025–26.01.2025,
 Final test: 27.01.2025–02.02.2025.

The reported metrics are ROC-AUC, Average Precision (AP), the Brier score, and positive-class lift in the top 10% of predictions. The main operational criterion is the daily AP difference between M2 and M1. Its uncertainty is estimated by resampling calendar days with replacement using 5,000 bootstrap replications. In the tables, “Brier improvement” is defined as $\text{Brier}_{M1} - \text{Brier}_{M2}$, so a positive value means a smaller squared probability error.

9 Results

Table 1 reports the mean daily M2-minus-M1 differences on the REAL-05A final test. For both target markets, all four metrics improve on every one of the seven test days.

Table 1: REAL-05A: mean daily M2-minus-M1 difference.

Market	Metric	Difference	95% interval
BTCUSDT	ROC-AUC	0.01452	[0.01238; 0.01696]
BTCUSDT	AP	0.01879	[0.01676; 0.02129]
BTCUSDT	Brier improvement	0.00301	[0.00210; 0.00388]
BTCUSDT	top-decile lift	0.06856	[0.05865; 0.07971]
ETHUSDT	ROC-AUC	0.00849	[0.00602; 0.01096]
ETHUSDT	AP	0.01120	[0.00892; 0.01385]
ETHUSDT	Brier improvement	0.00100	[0.00050; 0.00155]
ETHUSDT	top-decile lift	0.04268	[0.03438; 0.05305]

Pooled AP increases from 0.63155 to 0.64967 for BTCUSDT and from 0.63675 to 0.64763 for ETHUSDT. The evidence supports the hypothesis that synchronous cross-market state carries additional predictive information about aggressor-aligned markout at the tested short horizon.

Result 9.1. Under the fixed protocol, M2 outperforms the one-second M1 model and is the best validated specification among M0–M3. This conclusion concerns the ranking of aggressor-aligned markout at a five-second horizon.

10 Limitations and reproducibility

M2 inherits every limitation of M1 and adds several more. An observational relationship between markets does not establish causal direction: both markets may respond to

the same outside factor, creating the appearance of information transfer where none exists. Market synchronization is imperfect because public trades have discrete timestamps and do not reproduce the actual data-receipt latency of a trading system. Time windows of equal duration may also contain different numbers of events in the target and related markets, limiting the comparability of scales.

Historical `aggTrades` do not contain the order book, fees, position ownership, or actual execution costs. Zero-filling inactive seconds is part of the field definition; a different filling rule would require a new feature version. The seven-day final period provides a strong internal comparison, but it does not cover every market regime.

Features from different markets enter the logit additively. The model has no terms of the form $\phi_A\phi_B$, so it cannot represent a joint same-direction signal as more dangerous than the sum of its separate effects. This is exactly what the `_no_j` suffix means. One can introduce an interaction term

$$J\phi_A\phi_B,$$

where J controls the cross-market interaction. M3 carries out that experiment.

References

- [1] L. R. Glosten, P. R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100, 1985.
- [2] D. Easley, M. M. López de Prado, M. O’Hara. Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493, 2012.
- [3] R. Cont, A. Kukanov, S. Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1), 47–88, 2014.
- [4] L. P. Kadanoff. Scaling laws for Ising models near T_c . *Physics Physique Fizika*, 2, 263–272, 1966.
- [5] K. G. Wilson. Renormalization group and critical phenomena. I. *Physical Review B*, 4, 3174–3183, 1971.
- [6] Binance. Spot REST API: Compressed/Aggregate Trades List. <https://developers.binance.com/en/docs/catalog/core-trading-spot-trading/api/rest-api/market>.
- [7] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830, 2011.
- [8] Scikit-learn developers. `SGDClassifier` documentation. https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.SGDClassifier.html.
- [9] B. Efron, R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.